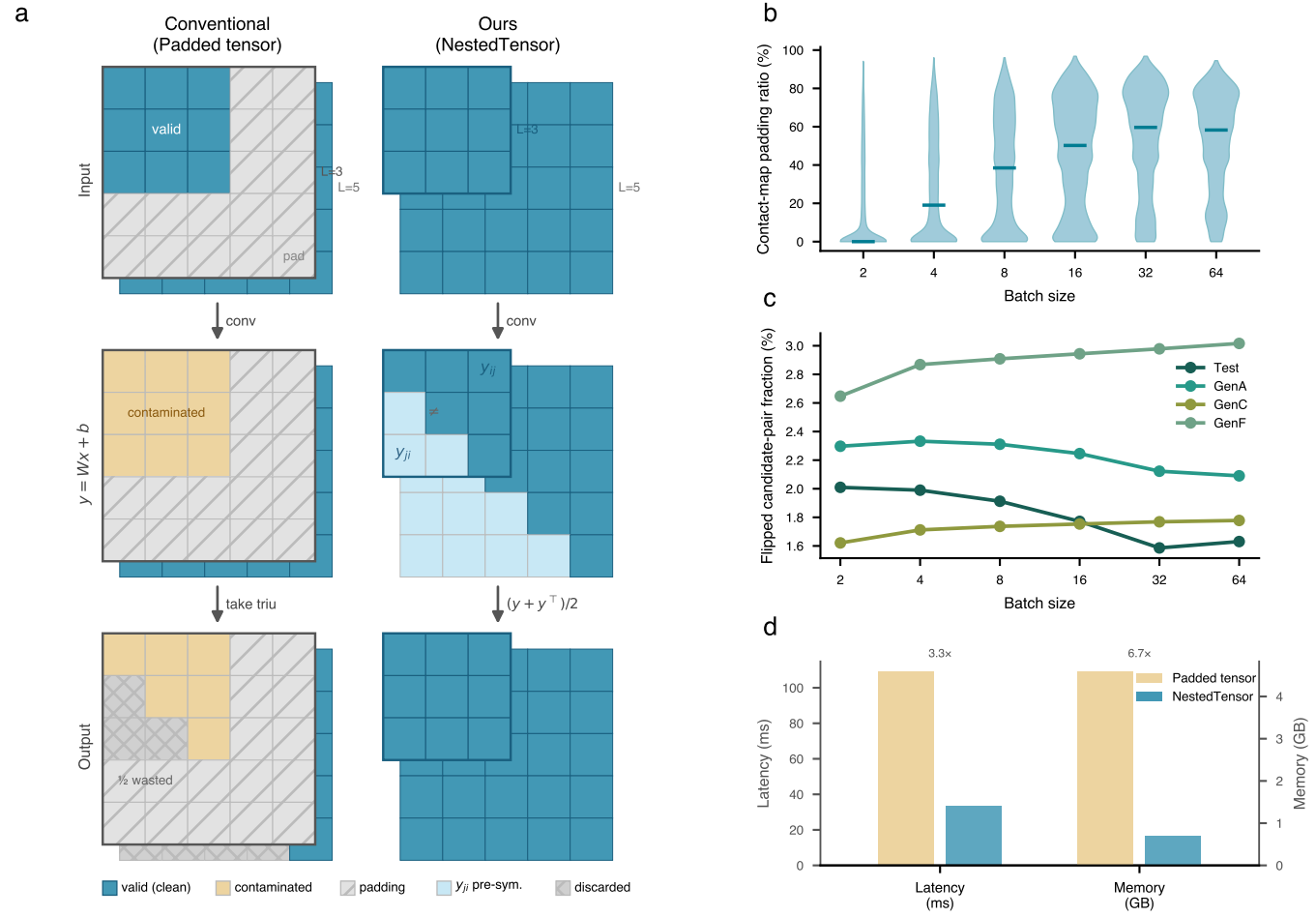
a
Conventional
(Padded tensor)
Ours
(NestedTensor)
Input
valid
L=3
L=5
pad
L=3
L=5
conv
conv
$y = Wx + b$
contaminated
y_{ij}
$\neq$
y_{ji}
take triu
$(y + y^\top)/2$
Output
½ wasted
valid (clean)
contaminated
padding
y_{ji} pre-sym.
discarded
b
Contact-map padding ratio (%)
100
80
60
40
20
0
2
4
8
16
32
64
Batch size
c
Flipped candidate-pair fraction (%)
3.0
2.8
2.6
2.4
2.2
2.0
1.8
1.6
Test
GenA
GenC
GenF
2
4
8
16
32
64
Batch size
d
3.3×
6.7×
Padded tensor
NestedTensor
Latency (ms)
100
80
60
40
20
0
Memory (GB)
4
3
2
1
0
Latency
(ms)
Memory
(GB)